

Obten todas las raíces de los siguientes complejos.

67) 2 raíces cuadradas de $4(\cos 30^\circ + i \sin 30^\circ)$

$$R = 4$$

$$\theta = 30^\circ$$

$$\sqrt[n]{r} \left(\cos \left(\frac{\theta + 360^\circ k}{n} \right) + i \sin \left(\frac{\theta + 360^\circ k}{n} \right) \right)$$

$$2\sqrt[2]{4} \left(\cos \left(\frac{30^\circ + 360^\circ \cdot 1}{2} \right) + i \sin \left(\frac{30^\circ + 360^\circ \cdot 1}{2} \right) \right)$$

$$2 \left(\cos \left(\frac{390^\circ}{2} \right) + i \sin \left(\frac{390^\circ}{2} \right) \right)$$

$$= 2(\cos 195^\circ + i \sin 195^\circ)$$

$$\star 2(\cos 195^\circ + i \sin 195^\circ)$$

$$2\sqrt[2]{4} \left(\cos \left(\frac{30^\circ + 360^\circ \cdot 0}{2} \right) + i \sin \left(\frac{30^\circ + 360^\circ \cdot 0}{2} \right) \right)$$

$$= 2 \left(\cos \left(\frac{30^\circ}{2} \right) + i \sin \left(\frac{30^\circ}{2} \right) \right)$$

$$\star = 2(\cos 15^\circ + i \sin 15^\circ)$$

68) 2 raíces cuadradas de $3(\cos 90^\circ + i \sin 90^\circ)$

$$\sqrt[2]{3 \left(\cos \left(\frac{90^\circ + 360^\circ \cdot 0}{2} \right) + i \sin \left(\frac{90^\circ + 360^\circ \cdot 0}{2} \right) \right)}$$

$$= \sqrt{3} \left(\cos \left(\frac{90^\circ}{2} \right) + i \sin \left(\frac{90^\circ}{2} \right) \right)$$

$$* = \sqrt{3} (\cos 45^\circ + i \sin 45^\circ)$$

$$\sqrt[2]{3 \left(\cos \left(\frac{90^\circ + 360^\circ \cdot 1}{2} \right) + i \sin \left(\frac{90^\circ + 360^\circ \cdot 1}{2} \right) \right)}$$

$$= \sqrt{3} \left(\cos \left(\frac{450^\circ}{2} \right) + i \sin \left(\frac{450^\circ}{2} \right) \right)$$

$$* = \sqrt{3} (\cos 225^\circ + i \sin 225^\circ)$$

69) $-4\sqrt{2} + 4i\sqrt{2}$

$$r = |z| = \sqrt{(-4\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{16 \cdot 2 + 16 \cdot 2}$$

$$\sqrt{32 + 32} = \sqrt{64} = 8$$

$$\tan \theta = \frac{4\sqrt{2}}{-4\sqrt{2}} = -1 = -45^\circ$$

$$-45^\circ + 180^\circ = 135^\circ$$

$$3\sqrt{8} \left(\cos \left(\frac{135^\circ + 360^\circ \cdot 0}{3} \right) + i \sin \left(\frac{135^\circ + 360^\circ \cdot 0}{3} \right) \right)$$

$$= 2 \left(\cos \left(\frac{135^\circ}{3} \right) + i \sin \left(\frac{135^\circ}{3} \right) \right)$$

$$* = 2 (\cos 45^\circ + i \sin 45^\circ)$$

$$3\sqrt{8} \left(\cos \left(\frac{135^\circ + 360^\circ \cdot 1}{3} \right) + i \sin \left(\frac{135^\circ + 360^\circ \cdot 1}{3} \right) \right)$$

$$= 2 \left(\cos \left(\frac{495^\circ}{3} \right) + i \sin \left(\frac{495^\circ}{3} \right) \right)$$

$$* = 2 (\cos 165^\circ + i \sin 165^\circ)$$

$$3\sqrt{8} \left(\cos \left(\frac{135^\circ + 360^\circ \cdot 2}{3} \right) + i \sin \left(\frac{135^\circ + 360^\circ \cdot 2}{3} \right) \right)$$

$$= 2 \left(\cos \left(\frac{855^\circ}{3} \right) + i \sin \left(\frac{855^\circ}{3} \right) \right)$$

$$= 2 \left(\cos \left(\frac{855^\circ}{3} \right) + i \sin \left(\frac{855^\circ}{3} \right) \right)$$

$$* = 2 (\cos 285^\circ + i \sin 285^\circ)$$

$$70. -\frac{27}{8}$$

parte 3 raices = $n=3 = 0, 1, 2.$

$$\sqrt[3]{\frac{27}{8}} \left(\cos\left(\frac{0^\circ + 360^\circ \cdot 0}{3}\right) + i \sin\left(\frac{0^\circ + 360^\circ \cdot 0}{3}\right) \right)$$

$$= \frac{3}{2} \left(\cos\left(\frac{0^\circ}{3}\right) + i \sin\left(\frac{0^\circ}{3}\right) \right)$$

$$(\cos = 0^\circ + \sin = 0^\circ)$$

$$\star = \frac{3}{2} (\cos 0^\circ + i \sin 0^\circ)$$

$$\sqrt[3]{\frac{27}{8}} \left(\cos\left(\frac{0^\circ + 360^\circ \cdot 1}{3}\right) + i \sin\left(\frac{0^\circ + 360^\circ \cdot 1}{3}\right) \right)$$

$$= \frac{3}{2} \left(\cos\left(\frac{360^\circ}{3}\right) + i \sin\left(\frac{360^\circ}{3}\right) \right)$$

$$= \frac{3}{2} (\cos 120^\circ + i \sin 120^\circ)$$

$$\star = \frac{3}{2} (-1 + 0i) = -\frac{3}{2}$$

$$\text{forma polar} = r = \frac{27}{8} (\cos \theta + i \sin \theta)$$

$$\sqrt[3]{27/8} = 3/2$$

$$\sqrt[3]{\frac{27}{8}} \left(\cos \left(\frac{0^\circ + 360^\circ \cdot 2}{3} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 2}{3} \right) \right)$$

$$= \frac{3}{2} \left(\cos \left(\frac{720^\circ}{3} \right) + i \sin \left(\frac{720^\circ}{3} \right) \right)$$

$$= \frac{3}{2} \left(\cos \left(\frac{720^\circ}{3} \right) + i \sin \left(\frac{720^\circ}{3} \right) \right)$$

$$\cancel{*} = \frac{3}{2} (\cos 240^\circ + i \sin 240^\circ)$$

$$71. -32i$$

$$\sqrt[5]{32} \left(\cos \left(\frac{270^\circ + 360^\circ \cdot 0}{5} \right) + i \sin \left(\frac{270^\circ + 360^\circ \cdot 0}{5} \right) \right)$$

$$= 2 \left(\cos \left(\frac{270^\circ}{5} \right) + i \sin \left(\frac{270^\circ}{5} \right) \right)$$

$$\cancel{*} = 2 (\cos 54^\circ + i \sin 54^\circ)$$

$$\sqrt[5]{32} \left(\cos \left(\frac{270^\circ + 360^\circ \cdot 1}{5} \right) + i \sin \left(\frac{270^\circ + 360^\circ \cdot 1}{5} \right) \right)$$

$$= 2 \left(\cos \left(\frac{630^\circ}{5} \right) + i \sin \left(\frac{630^\circ}{5} \right) \right)$$

$$\cancel{*} = 2 (\cos 126^\circ + i \sin 126^\circ)$$

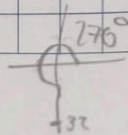
$$\begin{aligned} & \sqrt[5]{32} \left(\cos \left(\frac{270^\circ + 360^\circ \cdot 2}{5} \right) + i \sin \left(\frac{270^\circ + 360^\circ \cdot 2}{5} \right) \right) \\ &= 2 \left(\cos \left(\frac{270^\circ + 720^\circ}{5} \right) + i \sin \left(\frac{270^\circ + 720^\circ}{5} \right) \right) \\ &= 2 \left(\cos \left(\frac{990^\circ}{5} \right) + i \sin \left(\frac{990^\circ}{5} \right) \right) \\ & * = 2 (\cos 198^\circ + i \sin 198^\circ) \end{aligned}$$

$$\begin{aligned} & \sqrt[5]{32} \left(\cos \left(\frac{270^\circ + 360^\circ \cdot 3}{5} \right) + i \sin \left(\frac{270^\circ + 360^\circ \cdot 3}{5} \right) \right) \\ &= 2 \left(\cos \left(\frac{270^\circ + 1080^\circ}{5} \right) + i \sin \left(\frac{270^\circ + 1080^\circ}{5} \right) \right) \\ &= 2 \left(\cos \left(\frac{1350^\circ}{5} \right) + i \sin \left(\frac{1350^\circ}{5} \right) \right) \\ & * = 2 (\cos 270^\circ + i \sin 270^\circ) \end{aligned}$$

-32

$$R = \sqrt{(-32^2) + (0^2)} = \sqrt{1024} = 32$$

$$\tan^{-1} \left(\frac{32}{0} \right) = 270^\circ$$



$$= \sqrt[5]{32} \left(\cos \left(\frac{270 + 360 \cdot 4}{5} \right) + i \sin \left(\frac{270 + 360 \cdot 4}{5} \right) \right)$$

$$= 2 \left(\cos \left(\frac{270 + 1440}{5} \right) + i \sin \left(\frac{270 + 1440}{5} \right) \right)$$

$$= 2 \left(\cos \left(\frac{1710}{5} \right) + i \sin \left(\frac{1710}{5} \right) \right)$$

$$* = 2 (\cos 342^\circ + i \sin 342^\circ)$$

72. 6 roots of 729

$$n = 6, R = 729, \theta = 0^\circ, k = 1, 2, 3, 4, 5, 0$$

$$\sqrt[6]{729} \left(\cos \left(\frac{0 + 360^\circ \cdot 0}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 0}{6} \right) \right)$$

$$= 3 \left(\cos \left(\frac{0^\circ}{6} \right) + i \sin \left(\frac{0^\circ}{6} \right) \right)$$

$$* = 3 (\cos 0^\circ + i \sin 0^\circ), 3(1 + 0i) = \underline{3}$$

$$\sqrt[6]{729} \left(\cos \left(\frac{0 + 360^\circ \cdot 1}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 1}{6} \right) \right)$$

$$= 3 \left(\cos \left(\frac{360^\circ}{6} \right) + i \sin \left(\frac{360^\circ}{6} \right) \right)$$

$$* = 3 (\cos 60^\circ + i \sin 60^\circ)$$

$$\tan^{-1} \left(\frac{0}{729} \right) = 0^\circ$$

$$\sqrt[6]{729} \left(\cos \left(\frac{0^\circ + 360^\circ \cdot 3}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 3}{6} \right) \right)$$

$$= 3 \left(\cos \left(\frac{1080^\circ}{6} \right) + i \sin \left(\frac{1080^\circ}{6} \right) \right)$$

$$\ast = 3 (\cos 180^\circ + i \sin 180^\circ)$$

$$= 3(-1 + 0i) = -3$$

$$\sqrt[6]{729} \left(\cos \left(\frac{0^\circ + 360^\circ \cdot 4}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 4}{6} \right) \right)$$

$$= 3 \left(\cos \left(\frac{1440^\circ}{6} \right) + i \sin \left(\frac{1440^\circ}{6} \right) \right)$$

$$\ast = 3 (\cos 240^\circ + i \sin 240^\circ)$$

$$\sqrt[6]{729} \left(\cos \left(\frac{0^\circ + 360^\circ \cdot 5}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 5}{6} \right) \right)$$

$$= 3 \left(\cos \left(\frac{1800^\circ}{6} \right) + i \sin \left(\frac{1800^\circ}{6} \right) \right)$$

$$\ast = 3 (\cos 300^\circ + i \sin 300^\circ)$$

$$3 \left(\cos \left(\frac{0^\circ + 360^\circ \cdot 2}{6} \right) + i \sin \left(\frac{0^\circ + 360^\circ \cdot 2}{6} \right) \right)$$

$$\ast 3 (\cos 120^\circ + i \sin 120^\circ)$$